

Examiner reports

Q1.

In part (a), most correctly read the graph and then correctly used the coordinates to give the required P and V values.

The majority seemed well practised in the first step for part (b), writing the equation as $\log_{10} P = \log_{10} C + d \log_{10} V$. Many progressed to a correct solution easily and completed the question. Others became confused about what was a log and what was not or failed to handle the negative values and negative gradient. Those who tried to use point A and another point to generate simultaneous equations usually became lost at some point in the algebra.

In part (c), most remembered to include the units, relating the mathematical model back to the physical situation behind it.

Q6.

This question which tested the laws of logarithms caused problems for approximately half the students. Although many correctly wrote $\log_a N - \log_a x$ as $\log_a \frac{N}{x}$, a common error was to incorrectly write it as $\frac{\log_a N}{\log_a x}$. The majority of students who reached $\log_a \frac{N}{x} = \frac{3}{2}$ went on to present a full correct solution although others could not eliminate the logarithm correctly or failed to correctly change the subject of the resulting formula to write x in terms of N and a .

Q7.

The sketch of the given exponential curve was usually drawn having the correct intercept at $y = 1$, but a significant minority of students did not show its correct behaviour for negative x values – the curve should level out close to the x -axis and certainly nearer to the x -axis than the line $y = 1$. Part (b) was usually answered correctly with almost all showing that they knew how to use logarithms to solve an equation of the form $a^x = b$. The positioning of the minus sign was crucial in part (c) and it was not uncommon to see the incorrect answer $f(x) = -9^x$.

Q8.

In this question, which tested logarithms, most students scored the mark in part (a) but not all could see its use at a later stage in part (b). Not unexpectedly, part (b) proved discriminating, with a majority of students only picking up one of the six marks, normally for writing $2 \log(x + 7)$ as $\log(x + 7)^2$. The remaining students, who correctly applied a

further law of logarithms to reach the stage $\frac{(x+7)^2}{x+5} = 3$, either stopped at that point or went on to find the correct quadratic equation. In the latter case, most went on to solve the equation correctly but the required concluding statement ‘only one value of x ’ was not always given and so the final mark was lost. Some students who recognised what was being asked for frequently obtained the correct quadratic equation, showed that $b^2 - 4ac =$

0 and then stated 'only one value of x '. for full marks.

Q9.

This question, which tested the 'exponentials and logarithms' section of the specification, resulted in a good spread of marks. Although there were many correct sketches of $y =$

$\frac{1}{2^x}$ seen in solutions to part (a), there were almost as many that were sketches similar to that of $y = 2^x$. Another common incorrect sketch was one which was obtained by reflecting $y = 2^x$ in the x -axis.

Part (b) was generally answered well, although some candidates failed to pick up the final mark because they either left their final answer in the form $-x = 0.322$ or did not give the value of x to at least the required degree of accuracy. Of those who were less successful, some displayed poor algebraic skills or believed incorrectly that cross multiplying led to an equation of the form $10^x = 4$.

Part (c) proved to be demanding, with most candidates failing to obtain the correct expression for y in terms of a and b , although the majority of candidates were awarded partial credit for applying log laws correctly. A typical misconception is illustrated by $2 \log_a b + 3 \log_a y = 5 \log_a by$. Those candidates who were the most successful usually applied no more than one log law (or used $\log_a a = 1$) per line of working to reach, for example, $\log_a b^2 y = 1$ and then stated $b^2 y = a$ before completing by rearranging to obtain $y = ab^{-2}$.

Q10.

Very few candidates scored full marks for this question.

Part (a) was well done with only a handful failing to recognise a stretch and most giving the correct direction and scale factor; most who were wrong in one of these were wrong in both. It was essential to use the correct term 'translate' for the other, so 'shift' did not earn marks and 'transformation' was inadequate. A few made an error in the vector but most were correct.

Most candidates were unsuccessful in answering part (b) of the question as their curve was either below the x -axis or continued into the second quadrant. Many placed the curve through e instead of $e + 1$, and the curvature, for $x < e + 1$, was often wrong.

Correct answers in part (c)(i) were sometimes spoiled by poor algebra. It was worrying to see the misunderstanding of the log function, with $\ln(x - e)$ expanded as $\ln x - \ln e$ and incorrect subsequent work once a correct answer had been found: $e^{-1} + e = 0$ was a common such response.

In part (c)(ii), although the first mark for $x \geq 2e$ was often earned (though $x \leq 2e$ was common), the second mark was only gained by a very few candidates, as the part of the inequality concerning e was generally omitted completely.

Q11.

The sketch of the given exponential curve was generally well-done although some

students did not display sufficient care in sketching the curve in the negative- x region. In part (b), the majority of students were awarded the mark for correctly eliminating one of the variables, but a significant number could not form a quadratic with any confidence. Those who recognised that 7^{2x} could be written as $(7^x)^2$ usually had no difficulty in proceeding to form and solve a correct quadratic equation in y and then to find the correct value for the y -coordinate of the point of intersection, although a small minority of these students did not give sufficient consideration to the rejection of the inappropriate negative solution. The usual errors seen are illustrated by the following: $7^x = 7^{2x} - 12 \Rightarrow 12 = 7^{2x} - 7^x \Rightarrow 12 = 7^x$ or $7^x = 7^{2x} - 12 \Rightarrow \log 7^x = \log 7^{2x} - \log 12$. There were, however, an encouraging number of completely correct solutions.

Q12.

Most candidates stated, or clearly indicated, the correct values for p , q and r in part (a) and appreciated that these answers could then be used in part (b). There were many correct solutions seen, although a number of candidates resorted to using logarithms rather than just the rules of powers. The most common error in applying laws of indices

was to express $2^{\frac{1}{2}} \times 2^x$ as $2^{\frac{x}{2}}$ rather than $2^{\frac{1}{2}+x}$ which led to the common wrong answer $x = -6$. For full marks in part (b), candidates were required to state the value of x explicitly rather than leave the last line of their working as $2^x = 2^{-3.5}$.

Q13.

Graphs were drawn correctly by the majority of candidates and the correct vector given for the translation. There are still too many candidates, though, who use alternative words for 'translation' such as "move" or "shift" which are not accepted.

In part (c)(i) the justification of the transformation of the index equation was more of a challenge even to the high grade candidates. A substantial number of candidates simply replaced 4^x with $(2^x)^2$ which gained no credit as it was basically what they had to show. More were successful in showing that $2^{x+2} = 4Y$ by inserting the intermediate stage $= 2^x \times 2^2$. The final part was only done well by better candidates with many misusing the laws of logs at the outset as illustrated by 'log $4^x - \log 2^{x+2} = \log 5$ '. Of those who did solve the printed quadratic equation in Y , equated the values to 2^x and solved for x , where possible, only a minority gave a conclusion for the "show that" that there was "only one real solution".

Q14.

Part (a) was well done.

In part (b) the majority recognised what was required and obtained the relevant quadratic even if their notation was sometimes a bit suspect. A few used a substitution for $\ln x$ which helped, but those who used x instead of, say Y , occasionally forgot to change back to earn the final marks. The occasional candidate changed $3\ln x$ to $\ln(x^3)$ which was not helpful, and after multiplication by $\ln x$ there were a few $(3\ln x)^2$ terms seen.

Q15.

Many candidates obtained the correct values for x in parts (a)(i) and (a)(ii). The more able candidates often presented a full correct solution when solving the logarithmic equation in

part (b) but a common error, seen in solutions from less able candidates, was to assume that $\log(n - 4) = \log n - \log 4$, which frequently resulted in a single value for n even though the question asked for the possible values of n .

Q16.

Candidates' sketches generally displayed a thorough understanding of the graph of $y = 2^x$. In part (b)(i) the trapezium rule was frequently applied correctly and a smaller proportion of candidates than normal failed to give their final answer to the stated degree of accuracy. Part (b)(ii) was almost always answered correctly. In part (c), the question asked for a description of a geometrical transformation. Although answers which gave two separate translations were condoned, the penalty for descriptions which involved two different transformations, for example stretch and translation, was severe. However many correct descriptions were presented, which indicates the continued improvement in candidates' performance on this topic. Part (d), as expected, proved to be demanding. The common wrong approach is illustrated by ' $\log y = \log 2^{x+k} + \log 3$ '. Better candidates applied the correct substitution and manipulation to reach $2^k = 5$, but a significant minority of these candidates, after taking logs and using the log law, left their answer in the form k

$$= \frac{\log 5}{\log 2}.$$

Q17.

Most candidates answered part (a)(i) correctly, although a surprising number seemed to think that $A \times 0 = A$.

Most candidates did part (a)(ii) convincingly, usually choosing to evaluate the exponential expression as 0.95 ... and solving for A to demonstrate its value is 60 to two significant figures. Those who did not actually solve for A or did not show A as 59.9...were penalised.

Most candidates started part (b)(i) well, showing they had interpreted the information given correctly. However many could not manipulate their resulting $\ln$ expression into the form required; many, either not knowing what was meant by " a and b are integers" or not taking notice of it, left their result in fractions. Many candidates 'lost' a minus sign during their working whilst some attempted to take the $\ln$ of a negative number.

In part (b)(ii), a few candidates were able to derive the result clearly and efficiently; others managed it after some working that was somewhat difficult to follow whilst others made an error such as dropping a minus sign but still claimed they had obtained the result.

However, many candidates gave up after finding $\frac{dA}{dt}$. A few candidates attempted to integrate the given differential equation, some successfully recovering the given expression for A , which was both acceptable and impressive.

Although many candidates had part (b)(iii) correct, many errors were also seen, quite a

common one being $4 \times 2 = \frac{1}{2}$. The other common error was to substitute $h = 13$, rather

than $\frac{dA}{dt} = 13$. It was notable that many candidates attempted part (b)(iii) before returning

to try and sort out part (b)(ii), which was sensible.

Q18.

- (a) (i) The first part of this question was reasonably answered with many candidates obtaining both marks. Where candidates obtained 1 mark it was because many ended up with the answer $1/(5x - 2)$.
- (ii) Again this part was very well answered with candidates obtaining both marks. The majority of candidates arrived at $K\cos 2x$ but $K = -2$ was a common error.
- (b) (i) This part was not answered very well. Many candidates lost a mark through using -0.693 instead of $\ln\left(\frac{1}{2}\right)$ and $f(x) \geq 0$ was a common response.
- (ii) Most candidates were able to do this part with the correct answers often seen. The main error was the omission of brackets around $5x - 2$ obtaining $\sin 2(\ln 5x - 2)$. The expression for $fg(x)$ was also often seen.
- (iii) For those candidates with a correct starting expression many went on to get full marks. Those candidates who used $\sin \ln (5x - 2)^2$ often lost an accuracy mark for not rejecting one of their answers. Most candidates obtained the first method mark for making the correct initial step for their expression.
- (iv) This was usually well done but there were common errors of dividing by $\sin 2$ obtaining $y/\sin 2 = x$ or even $y/\sin = 2x$ and $y/2 = \sin x$

Q19.

Most candidates were able to score marks in part (a), although the mark in part (a)(iii) was not awarded for answers left as ' $\log_e 5^3$ '. In part (b), candidates generally scored the method marks but frequently failed to pick up the accuracy mark. The most common wrong answer was '1.66', not always simply from a premature approximation.

Part (c), as expected, proved to be beyond the ability of the average candidate; some did score a mark for ' $p = 2^m$ ', but this was often followed by ' $pq = 2^m \times 8^n = 16^{mn}$ '. Better candidates reached ' $pq = 2^m \times 2^{3n}$ ', but a significant number of these candidates then simplified incorrectly to reach the wrong answer ' $pq = 2^{3mn}$ '.

Q20.

Part (a) was well answered, with most candidates gaining both marks. The most common

error was $3 \ln e^x = \ln 4$, resulting in $x = \frac{\ln 4}{3}$.

Part (b) was generally well done.

Part (c) was not very well answered. Most candidates obtained 1 mark for the y -values but many candidates stopped at this stage and therefore lost the final 2 marks. The error in part (a) was often repeated.

Q21.

The work required to answer parts (a)(i) and (a)(ii) was well understood but a significant number of candidates did not give an explicit value for p or an explicit value for x . Candidates at this level should not be leaving their answers in an embedded form, for example, in part (a)(ii) answers were sometimes left as ' $52 \times 0.75 = 125$ '.

Part (b) was answered better than the corresponding work in previous years and there continued to be an improvement in knowledge and use of the laws of logarithms as seen in solutions to part (c) where many candidates were able to score at least half marks by correctly applying two laws of logarithms. Only more able candidates either used $\log_a a = 1$

or applied $\log_a \frac{x}{k} = -1 \Rightarrow \frac{x}{k} = a^{-1}$ towards expressing x in terms of a correctly.

Q22.

This question proved to be the downfall for many of the candidates. It was the poorest answered question on the paper.

Many candidates answered part (a) correctly with the correct answer often seen. A common error was not to give the answer in an exact form, but this only affected candidates who showed no working. The solution $x = \ln(5/2)$ was also common.

In part (b) the only candidates who were really successful here were the ones who used a substitution such as $y = \ln x$, who then were able to formulate and solve the quadratic very easily. Those candidates who attempted to work in $\ln x$ usually (despite condoning poor notation) failed to obtain a quadratic and hence scored zero. Many candidates just substituted $2\ln x = 5$ and verified that it fitted the equation.

Q23.

Many candidates gave a correct solution for part (a) but only a minority gave full correct solutions to part (b). Average grade candidates in general obtained the method marks for applying two logarithmic laws correctly but only the more able candidates could deal with the constant 1 correctly. Instead of writing 1 as $\log_a a$, the common wrong approach was to write $\log_a 36 + 1$ as $\log_a 37$.

Q24.

In part (a), many candidates gave the correct value for where the curve met the y -axis, but a significant minority drew no part of the graph in the second quadrant, their graphs stopping at the point (0, 1). Many candidates scored at least half marks for their descriptions of the geometrical transformations in part (b). The most common wrong answer in part (b)(i) was "a stretch of scale factor 2 parallel to the y -axis", and in part (b)(ii) the direction of the translation vector caused the most problems.

Part (c)(i) was poorly answered, with many candidates failing to give sufficient justification for the printed answer. A very common error was to write 9^x as $3y$ and yet still reach the correct factorised quadratic equation.

Although part (c)(ii) was answered better, the majority of candidates failed to gain all the four marks. The most common wrong approach was to take logarithms of each term in the given equation. Those candidates applying this incorrect method had clearly not seen the

implication of the “Hence”. Those candidates who had seen the link and started with $3^x = 1$ and $3^x = 2$ generally scored all four marks.

Q25.

This question fell into two clear parts: most candidates gained full, or nearly full, marks on part (a), but most were only able to score the first mark in part (b). In part (a)(i), most wrote down the value of A correctly. In part (a)(ii), most candidates obtained an expression for k from which it could be evaluated and so gained the marks. Those who did not give an explicit expression for k lost a mark, and those who just verified that $20k^{60} \approx 2000$ scored no marks. Similarly, candidates were mostly correct in part (a)(iii), but a few made an error in $2008 - 1885$, and some did not round to the nearest 1000.

In part (b), many candidates showed that they had understood the question by setting up the correct equation, but few of those were then able to solve it correctly. The common problem was that, having written down an equation of the form $ak^t = bm^t$, candidates wrote the corresponding log equation as $t \log(ak) = t \log(bm)$ and so, confused by their t apparently cancelling, abandoned their answer. Those who wrote the log equation correctly usually went on to complete the question correctly. The other common mistake was a misunderstanding of the question in which the expression for Q was equated to the result from part (a)(iii); this scored no marks.

Q26.

This question, which tested the basic definition of a logarithm as well as testing the laws of logarithms, was answered much better than in previous series. The most common error

was in part (b), where $\log_a 30 - \log_a 15 = \frac{\log_a 30}{\log_a 15}$ was seen quite often.

Q27.

Candidates generally applied the trapezium rule correctly and gave the final answer to the given degree of accuracy. Many candidates failed to gain full marks in part (b)(ii) as their diagrams did not show four trapeziums. There were also a significant number of candidates who either failed to draw any diagram or who drew the sloping sides of their trapeziums under the curve. Most candidates realised that the geometrical transformation was a stretch, but the wrong answer ‘stretch parallel to the y -axis, scale factor 3’ appeared almost as frequently as the correct answer. A high proportion of candidates showed that they were able to use logarithms to solve the equation $6^{3x} = 84$ but some only gave their answer to two decimal places. The final part of the question was poorly answered, although a minority of candidates did score some credit for partially correct answers, for example, $f(x) = 6^{x+1} - 2$.

Q28.

Part (a) was usually B1B0 with the shape being correct but the coordinates of the intercepts on the axes were often incorrect.

In part (b)(i) many fully correct answers were seen and most candidates earned partial credit. Mistakes usually occurred where candidates started with the translation and gave the combined vector. Where candidates started with the stretch they were far more

successful.

Part (b)(ii) was very well answered with many candidates earning full marks.

Q29.

Most solutions to part (a) were correct but some weaker candidates resorted to using 10 for a or just stated " $x = 8/3$ ". The majority of candidates gained credit for applying a correct law of logarithms in part (b) but many then made the error of writing $\log_a 6^3 - \log_a 8$

as
$$\frac{\log_a 6^3}{\log_a 8}$$

Hence, full credit was not awarded to those candidates who went directly from $\log_a 6^3 - \log_a 8$ to the printed answer: $\log_a 27$. In part (c)(i), a significant minority failed to introduce

p and left their final answer as " $y = \log_{10} \left(\frac{27}{8} \right)$ ". The final part of this question was frequently not attempted. Although some very good solutions were presented, it was surprising to see some otherwise able candidates quoting wrong formulae; " $\text{gradient} = (x_2 - x_1) \div (y_2 - y_1)$ " was the most common of these.

Q30.

This question also proved to be a good source of marks for a majority of candidates. Normally part (a) was answered correctly but there were also signs of poor examination technique. Candidates who just wrote down the y -coordinate of P as '3' scored no marks but those who wrote ' $3(2^0 + 1) = 3$ ' scored 1 of the 2 marks available. Responses to the trapezium rule question were mixed with many candidates scoring full marks but a greater number of candidates failing to pick up even the method mark. The two main method errors involved either the use of seven ordinates rather than the required four or the use of $x = 0, 1, 2, 3$ only.

In part (c)(i) a significant number of candidates were unable to gain the mark because they wrote 3×2^x as 6^x . Many candidates gained the 2 method marks for using logarithmic rules but a significant minority failed to round 2.5849 correctly to three significant figures. The wrong answer 2.59 was seen more than expected.

Q31.

Many candidates correctly used the laws of logarithms to write $\log_a 3(2n - 1)$ but then made an error in expanding the brackets either before or after eliminating the logarithms. A surprising number of candidates reached the correct equation ' $n = 6n - 3$ ' but were then unable to solve it correctly. Some candidates rearranged the given equation to have 0 on one side but these candidates often made an error in their subsequent work when trying to eliminate the logarithm. Weaker candidates started by writing ' $\log_a n = \log_a 3 + \log_a 2n - \log_a 1$ ' and so scored no marks for part (a).

Those candidates who attempted part (b)(i) generally obtained the correct expression, although the usual wrong answers such as 3^a or $\log 3_a$ were still seen.

In part (b)(ii) a significant number of above average candidates scored the first two marks

for ' $\log_a(y/8) = 4$ ', but, surprisingly, some candidates who had answered part (b)(i) correctly could not eliminate the logarithm to find y in terms of a . Those candidates who tried to multiply by x or $\log x$ at an earlier stage often got themselves into a mass of incorrect algebraic work.

Q32.

The evaluations in part (a) were done correctly by most candidates, although some made numerical errors. In part (a)(ii), the request for three significant figures was largely adhered to, although some interpreted this as three decimal places.

Candidates started part (b) well but many were confused as to how to handle the negative signs.

Many attempts to take logs of negative numbers were seen. Many got to $14 \ln\left(\frac{5}{12}\right)$ or $-14 \ln\left(\frac{5}{12}\right)$ and gave this as their answer. Despite this, the vast majority gave the correct answer of 12 to part (b)(ii), some just ignoring their negative sign or saying it must be positive. They apparently did not respond to a negative answer by reviewing their answer to part (b)(i). Very few candidates explained what they were doing with the negative index to obtain the expected answer. In part (c)(i), very few candidates derived the differential equation with confidence. Some made a start by attempting to differentiate the expression

for x , although, of those, many made the error of bring down the t together with $\frac{1}{14}$. Others attempted to find an expression for t and differentiate that. There was usually little progress beyond these starting points, although several candidates attempted to convince the examiner that they had found the requested result, with much abuse of the $\ln$ function seen in such attempts.

Some tried to integrate the given differential equation to recover the expression in x , and some very neat solutions were seen this way, although those who did not consider limits or a constant could make no further progress. Again, handling negative signs with logarithmic expressions caused difficulties. Most candidates scored the final mark for

applying the result from part (c)(i), although some did say that $\frac{1}{14}(7) = 2$.

Q33.

Although this question on logarithms was answered better than corresponding questions in summer 2005, the topic continues to be one which candidates find difficult. Many candidates gained credit for writing ' $2\log n = \log n^2$ ' but then often made the error ' $\log(5n - 24) = \log 5n - \log 24$ '. In part (b) it was disappointing to find so many candidates unable to solve the quadratic equation. It was not uncommon to find solutions which involved the wrong factors of the form $(n \pm 24)(n \pm 4)$. In general, only the better candidates gave a fully correct solution for both parts of this question.

Q34.

(a) The correct value of a was often seen, but $a = -9$ was a common incorrect answer.

The majority of candidates gained the method mark in finding b, but a number of candidates then continued with incorrect manipulation of the exponential function.

- (b) The majority of candidates answered this correctly.

Q35.

- (a) Reasonable attempts were made by most candidates. The main mistake was finding the derivative of x^{-2} which was often seen as $-2x^{-1}$. This lost marks since candidates then 'found' the answer given by incorrect means.
- (b) There were many fully correct answers but many candidates chose incorrect functions for u and $\frac{dv}{dx}$, with the obvious implications.
- (c) (i) This was reasonably well answered, the main error seen being $1 - 21nx = x^3$.
- (ii) The majority of candidates obtained the method mark following through from their answer to part (b). Few candidates completed this question correctly.



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.